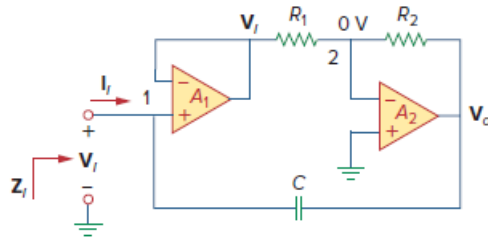


Problem

Determine the equivalent capacitance of the op amp circuit in Fig. 10.41 if $R_1 = 10 \text{ k}\Omega$, $R_2 = 10 \text{ M}\Omega$, and $C = 10 \text{ nF}$.

Figure 10.41: Capacitance multiplier.



Step-by-step solution

Step 1 of 4

Refer to Figure 10.41 in the textbook.

Write the expression for input impedance, (Z_i) .

$$Z_i = \frac{V_i}{I_i}$$

Write the expression for current, I_i .

$$I_i = \frac{V_i - V_0}{\frac{1}{j\omega(10 \times 10^{-9})}}$$

$$I_i = (V_i - V_0) \left(j\omega(10 \times 10^{-9}) \right) \dots\dots (1)$$

Step 2 of 4

Apply Kirchhoff's current law at node 2.

$$\frac{0 - V_i}{10 \times 10^3} + \frac{0 - V_0}{10 \times 10^6} = 0$$

$$\frac{(-V_i)(1 \times 10^3) - V_0}{10 \times 10^6} = 0$$

$$V_0 = -(V_i)(1 \times 10^3) \dots\dots (2)$$

Step 3 of 4

Substitute equation (2) in equation (1).

$$I_i = \left[V_i - (-V_i)(1 \times 10^3) \right] \left[j\omega(10 \times 10^{-9}) \right]$$

$$= V_i (1 + 1000) \left(j\omega(10 \times 10^{-9}) \right)$$

$$\frac{V_i}{I_i} = \frac{1}{j\omega(10 \times 10^{-9})(1001)}$$

Determine the input impedance.

$$Z_i = \frac{V_i}{I_i}$$

$$Z_i = \frac{1}{j\omega(10.01 \times 10^{-6})} \dots\dots (3)$$

Step 4 of 4

Write the general expression for input impedance of equivalent capacitance.

$$Z_i = \frac{1}{j\omega C_{eq}} \dots\dots (4)$$

Compare equation (3) and equation (4).

$$C_{eq} = 10.01 \times 10^{-6}$$

$$\cong 10 \times 10^{-6}$$

$$\cong 10 \mu\text{F}$$

Thus, the equivalent capacitance, C_{eq} is $10 \mu\text{F}$.